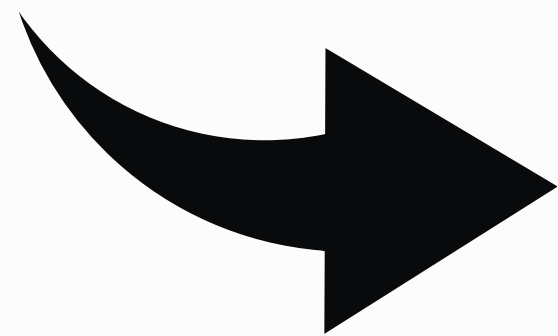


EVALUATION METRICS

IN
RAG



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What are evaluation metrics?

Evaluation metrics are ways to measure how good your RAG system actually is.

A RAG system has 2 brains:

1. Retriever → Finds relevant chunks
2. Generator (LLM) → Uses those chunks to answer

Metrics tell us:

- Did we retrieve the right information?
- Did the model use it correctly?
- Did it hallucinate or stay grounded?

Why we need them

- To reduce hallucinations
- To improve search quality
- To compare different RAG setups
- To move from “it sounds right”
→ “it is measured”



Types of RAG Evaluation Metrics

RAG evaluation happens at two levels:

1. Retrieval-Level Metrics

Measure how good the retriever is at finding useful chunks.

- Recall@K
- Precision@K
- Context Relevancy

2. Generation-Level Metrics

Measure how good the LLM's answer is.

- Faithfulness
- Answer Relevancy
- Groundedness

Retrieval-Level Metrics



- Recall@K
- Precision-KK
- Context Relevancy

Generation-Level Metrics



- Faithfulness
- Answer Relevancy
- Groundedness

Recall@K

What is it?

Measures whether the correct chunk is present in the top K retrieved results.

👉 “Did we at least fetch the right info?”

Formula (conceptually):

$$\text{Recall@K} = \frac{\text{Relevant chunks retrieved}}{\text{Total relevant chunks}}$$

Example

Question:

"What is the capital of France?"

Chunks in database

- C1: Paris is capital of France ✓
- C2: France population data
- C3: Berlin is capital of Germany

Retriever returns Top 2 (K=2) → C2, C1

Relevant chunk = C1

It is in top 2 → Recall@2 = 1 (Perfect)

If C1 was not retrieved → Recall@2 = 0

● High recall = system doesn't miss important info

Precision@K

What is it?

Measures how many of the retrieved chunks are actually useful.

👉 “Did we retrieve mostly relevant stuff or junk?”

Formula (conceptually):

$$\text{Precision@}K = \frac{\text{Relevant retrieved chunks}}{K}$$

Example

Same question: capital of France

Retriever returns Top 3:

- C1: Paris is capital of France ✓
- C2: France population ✗
- C3: Eiffel Tower tourism ✗

Only 1 relevant chunk.

$$\text{Precision@}3 = 1 / 3 = 0.33$$

● High precision = cleaner context

● Low precision = noisy context, LLM may get confused

Context Relevancy

What is it?

Measures how relevant the retrieved context is to the question.

Not just “correct chunk present”

But “How strongly related is the context?”

Usually scored by an LLM or similarity model.

Example

Question: “Side effects of paracetamol?”

Retrieved:

- C1: Paracetamol side effects list ✓
- C2: How to store medicines ✗

Even if answer can be generated, C2 lowers context relevancy score.

- High context relevancy → focused knowledge
- Low → LLM forced to ignore noise

Faithfulness

What is it?

Checks whether the answer is supported by retrieved context.

👉 Detects hallucinations.

Formula (conceptually):

Faithfulness=Is every claim backed by context?

Example

Context chunk:

“Paris is capital of France.”

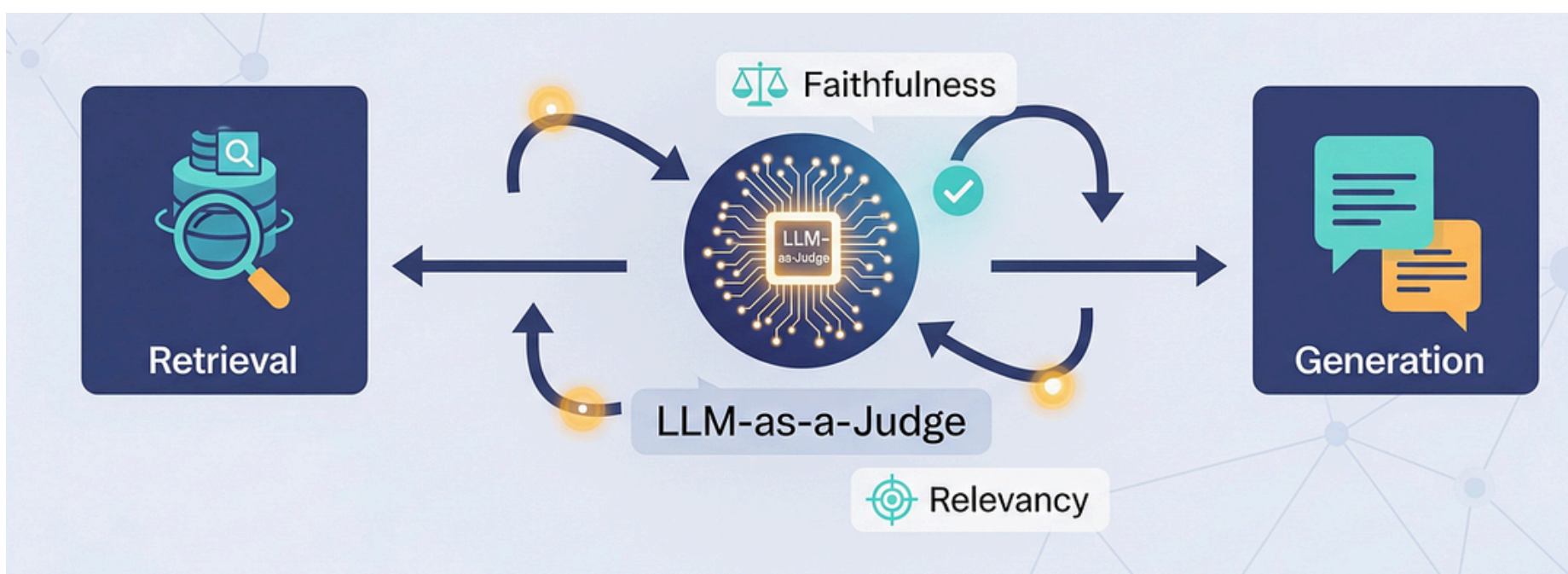
Model answer:

“Paris is capital of France and has population 5 million.”

Population not in context ❌ → Unfaithful

● Faithful = grounded in sources

● Unfaithful = hallucinated additions



Groundedness

What is it?

Measures how strongly the answer is tied to provided documents.

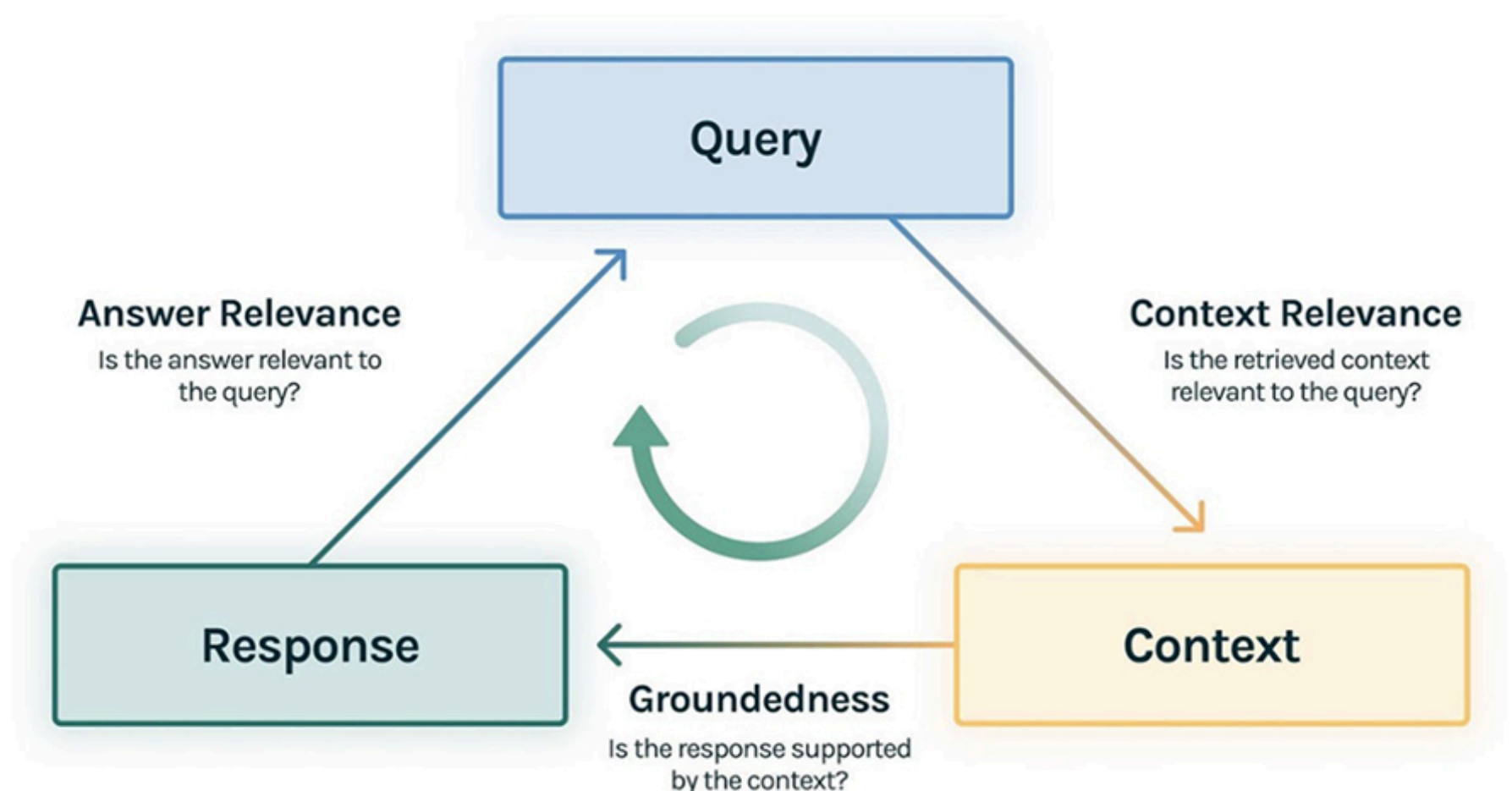
Similar to faithfulness but more about source dependence.

If the model could answer without context → low groundedness.

Example

Context: Indian tax rules

Answer: Generic tax advice not from context → poorly grounded



Answer Relevancy

What is it?

Measures how well the answer addresses the user's question.

Retriever might work fine, but model might go off-topic.

Example

Question: "Symptoms of diabetes"

Answer:

"Diabetes is a metabolic disorder discovered in 1500 BC."

Factually correct but not answering symptoms →
Low relevancy

● High relevancy = directly answers question

Tools for RAG Evaluation

You don't need to do all this manually.

Popular tools:

- Ragas → Specialized RAG metrics (faithfulness, context relevance)
- TruLens → Tracks grounding + feedback
- DeepEval → LLM-based scoring
- LangChain → Evaluation chains
- LlamaIndex → Built-in eval modules



RAG-Specific Metrics



TruLens

Grounding & Feedback



DeepEval

LLM-Based Scoring



LangChain

Evaluation Chains



LlamaIndex

Built-In Eval Modules